Weekly Report

Clustering Identification and Analysis Work

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TRIUMF DarkLight Group Meeting

August 18, 2026

What I’ve been working on

1. Cluster Identification
2. Cluster Analysis

Cluster Identification

Tests with Run 4010, Event 5 (elastic)

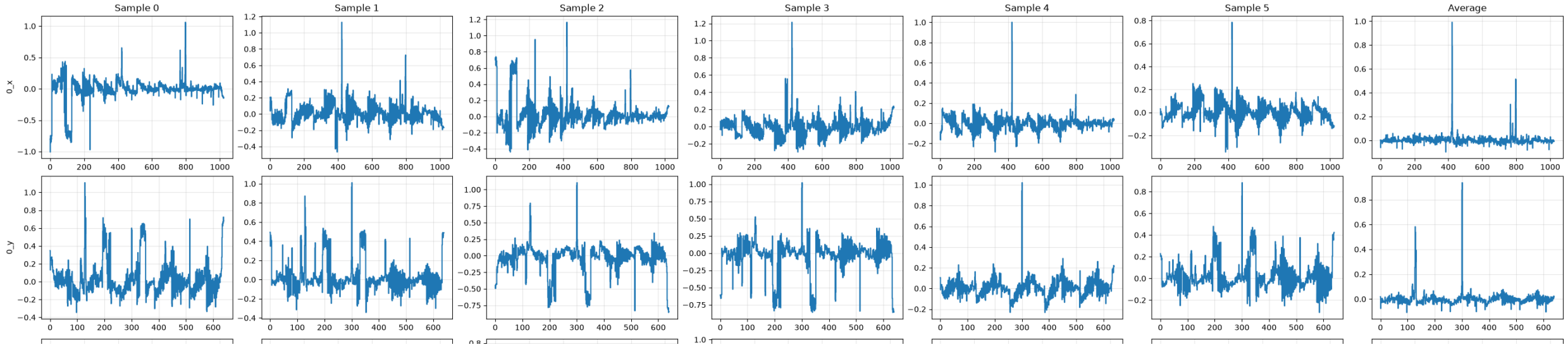
- Time-sample averaging (samples 3-5 versus all samples)
- Channel smoothing (Savitzky-Golay, moving average, triangular and Gaussian kernels)
- Second derivative threshold peak detection
- Gaussian derivative matched filter
- “Gaussian quality”

Preferred cluster identification algorithm (based on initial tests)

1. Average over all time samples
2. Channel smoothing with 5–channel Gaussian kernel
3. Peak detection in -y” multiplied by y (averaged, smoothed, and zeroed)
    - Threshold of 5 standard deviations above noise (all values < 99th percentile)

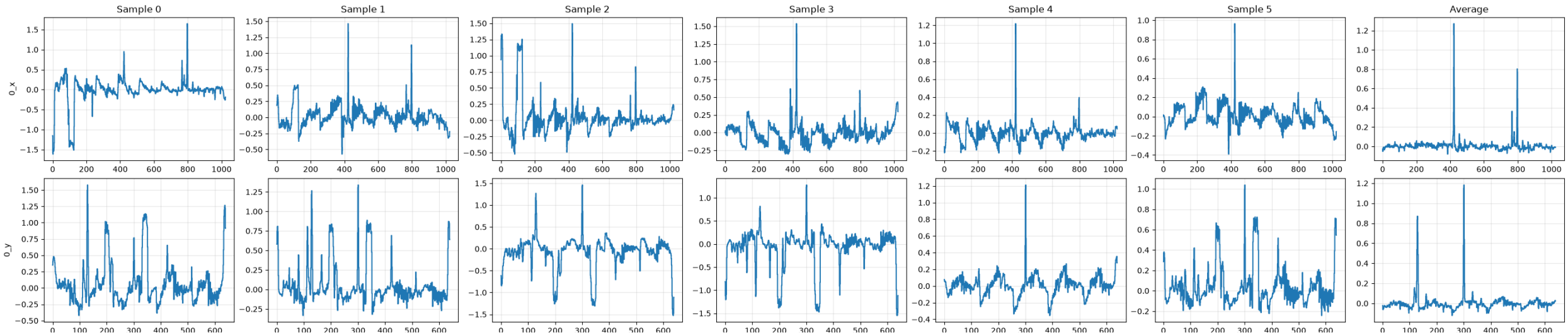
Pedestal and common-mode corrected GEM data

Left bottom GEM
Run 4010, Event 5

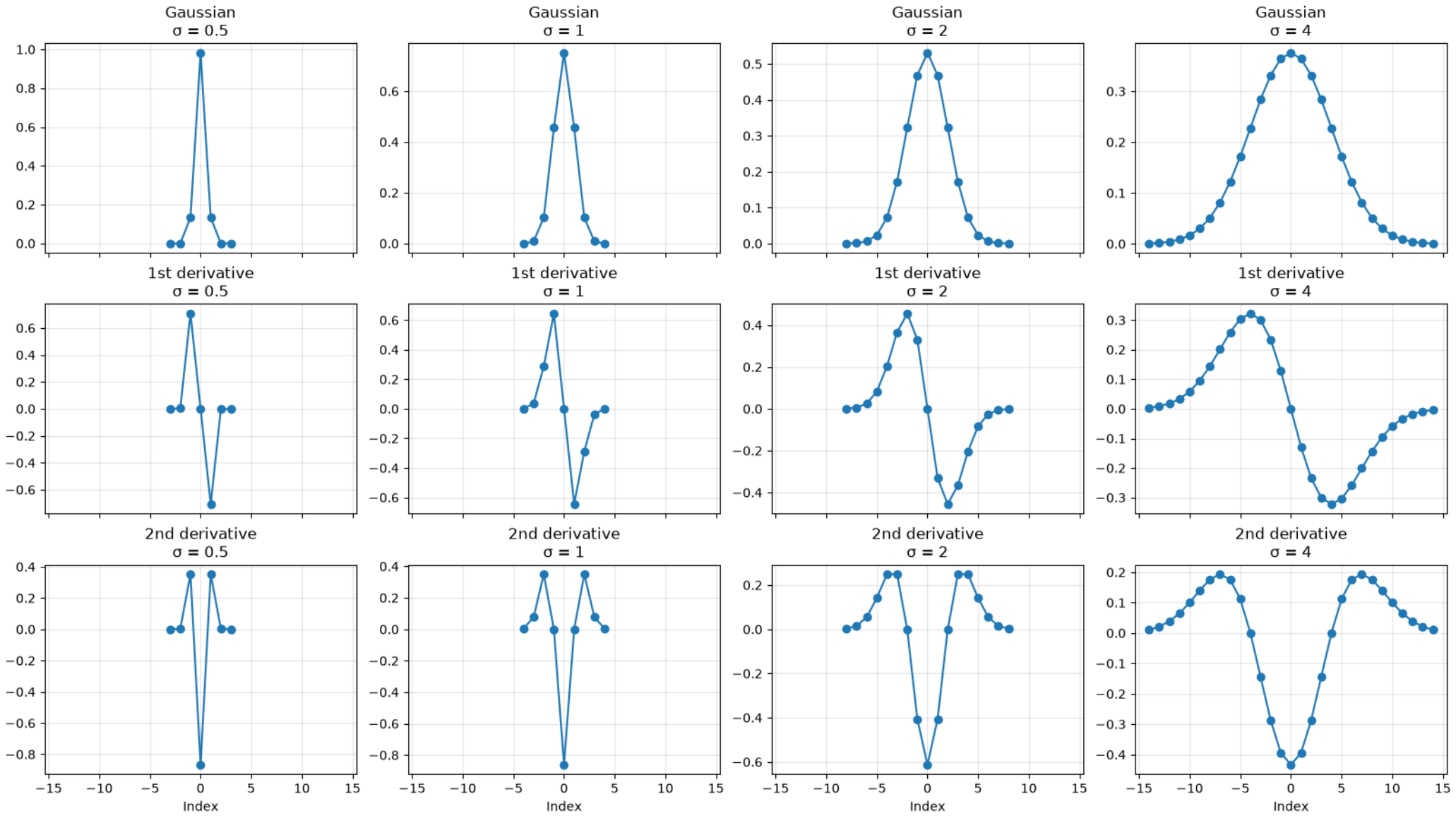


After applying smoothing

Smoothed with Gaussian template (sigma=1 channel, 9 channels wide)



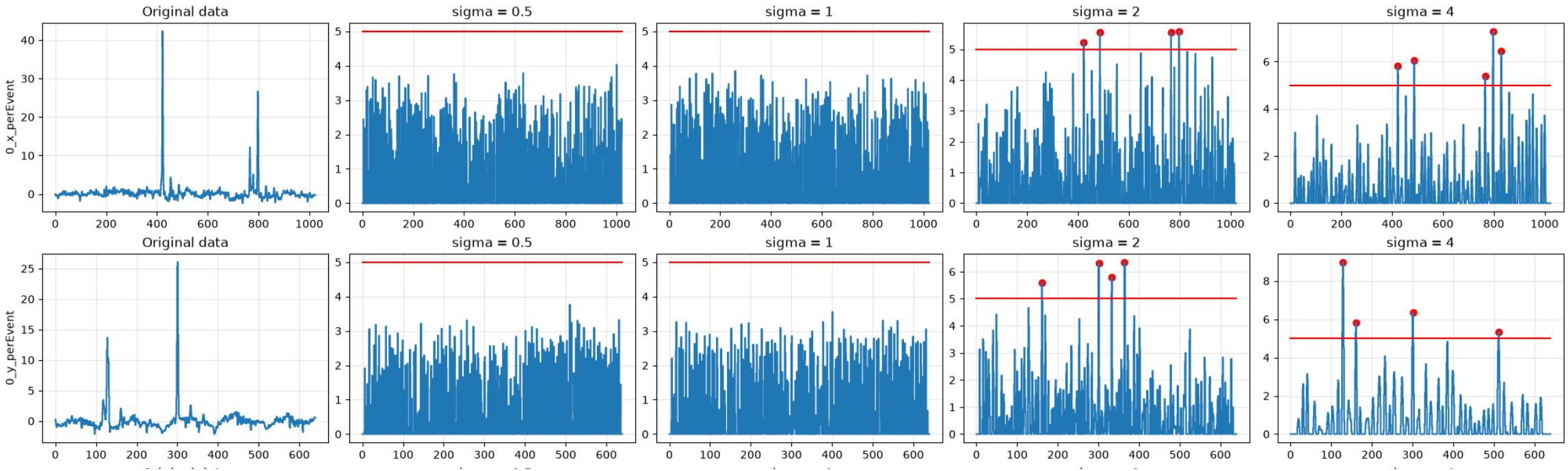
Gaussian templates



Second derivative matched filter

Convolute second derivative of smoothed data (y”) with second derivative gaussian templates

Scale = Signal / σ_{noise}



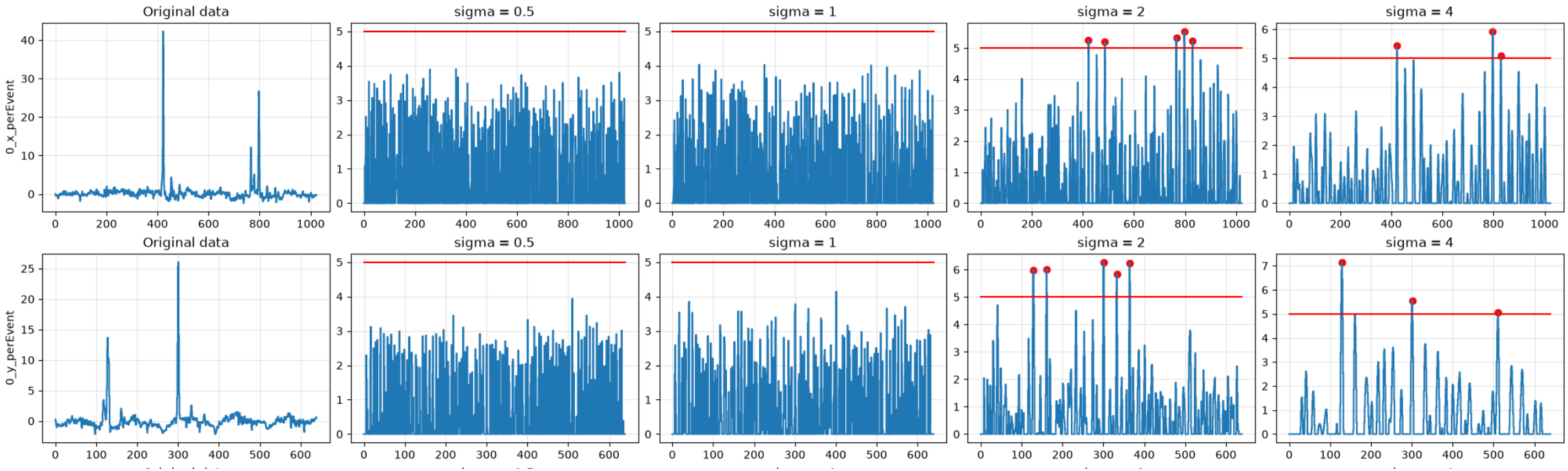
5 standard deviation threshold

(Negative outputs removed because they represent a match with reverse template)

First derivative matched filter

Convolute first derivative of smoothed data (y') with first derivative gaussian templates

Scale = Signal / σ_{noise}

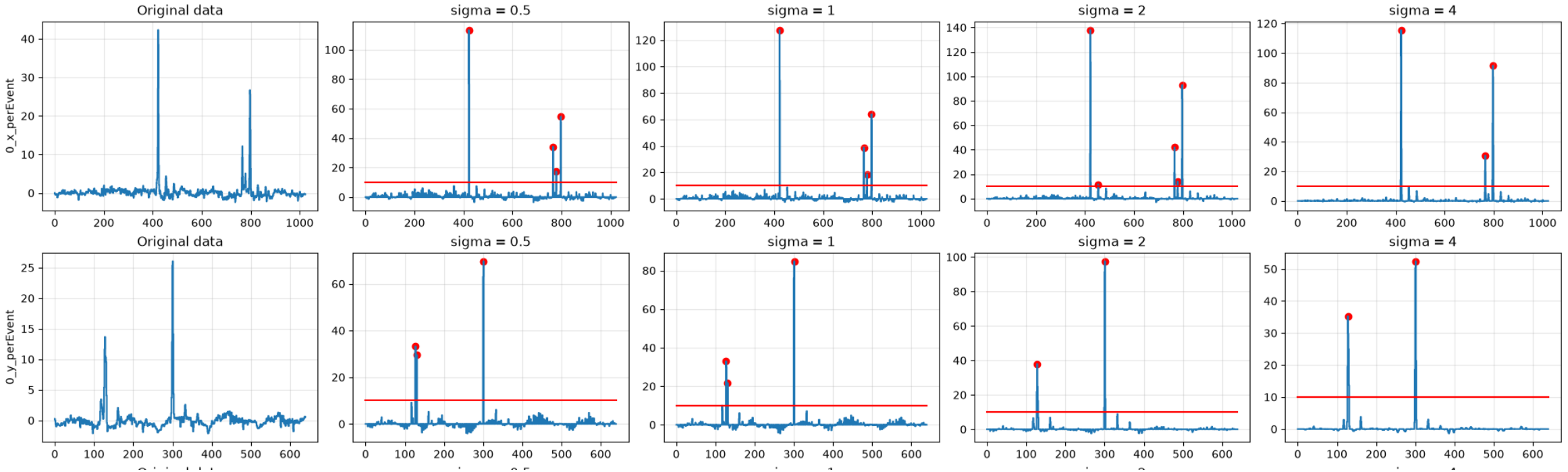


5 standard deviation threshold

Second derivative matched filter (enhanced)

Multiply smoothed data with output of convolution

Scale = Signal / σ_{noise}

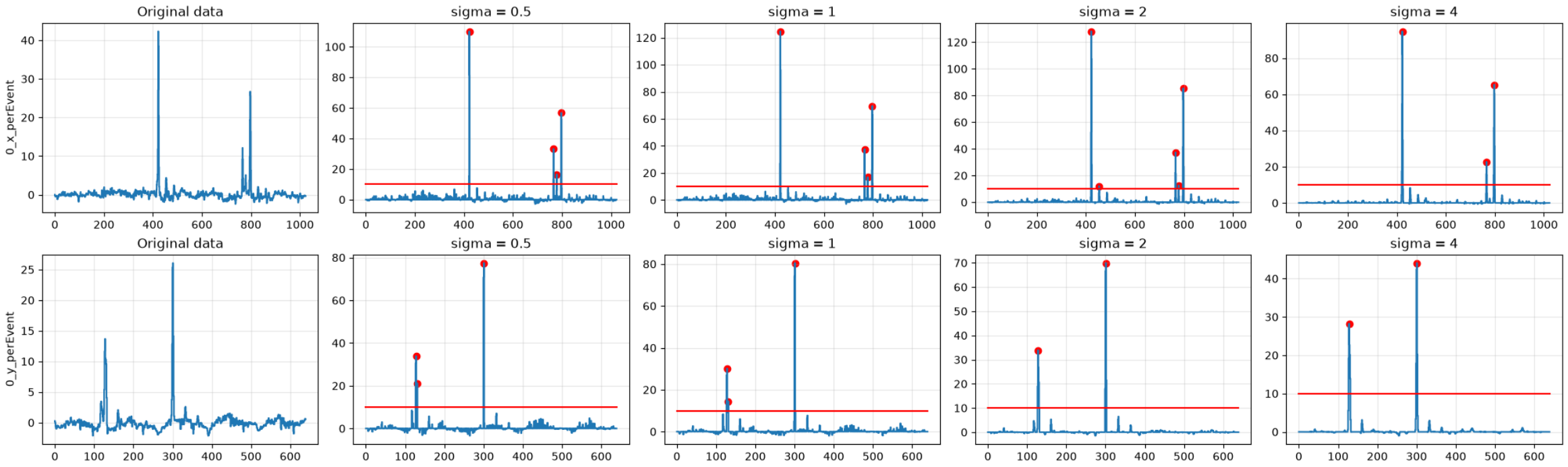


10 standard deviation threshold

First derivative matched filter (enhanced)

Multiply smoothed data with output of convolution

Scale = Signal / σ_{noise}



10 standard deviation threshold

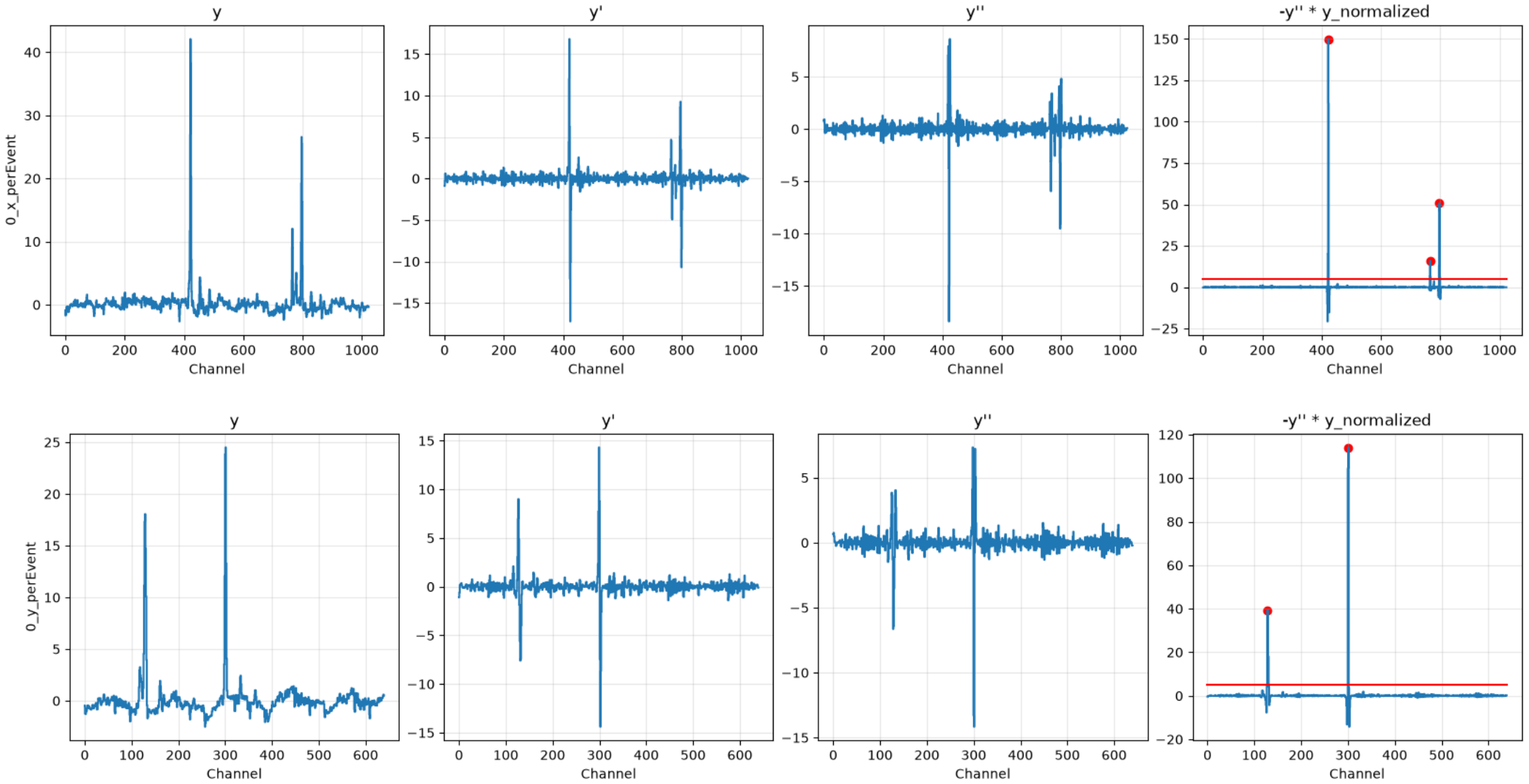
Thoughts on using a matched filter

- Matched filter output does not do a great job of resolving cluster peaks on its own, but it does improve SNR when multiplied with original data
- Second derivative filter produces slightly higher SNR
- First derivative is less sensitive to noise and requires less computation
- Continuous wavelet transform
  - Apply convolution with range of template sizes
  - Takes too much time
- Real clusters most similar to sigma = 0.5 – 1 templates

Derivative-based peak detection

Computationally simpler. Similar SNR gain to matched filter although perhaps not as good at enhancing buried signals

Scale = Signal / σ_{noise}



5 standard deviation threshold

Cluster Identification

Next Steps

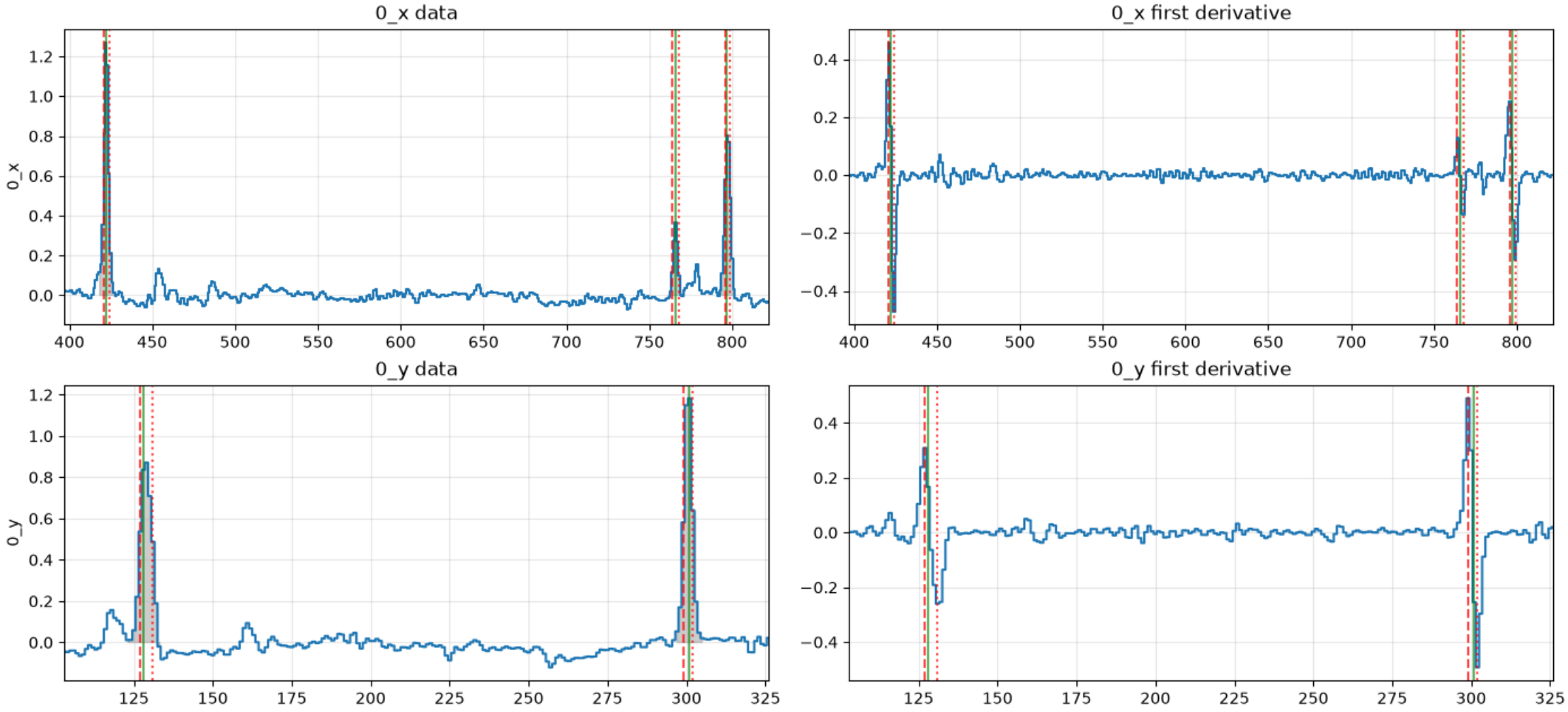
- Refactor code
  - Modularity – allow for different versions of cluster identification and analysis functions
  - Tunability – put thresholds in cutsDB framework?
- Test this new method with larger sample size
  - Try varying smoothing and threshold parameters
  - Compare against old method
- Measure matching efficiency

Cluster Analysis

- Estimating cluster properties
  - Width
  - Peak charge
  - Total charge
  - Position
  - Time
- Matching X and Y clusters
  - Total charge as a function of time
  - Charge amplitude?
- Angle of incidence
  - Time of maximum charge versus channel

Width calculation

1. Estimate σ from nearby extrema in first derivative
2. Calculate charge within $\pm 2.756\,\sigma$ of peak (~99% of area)



Charge over time profiles

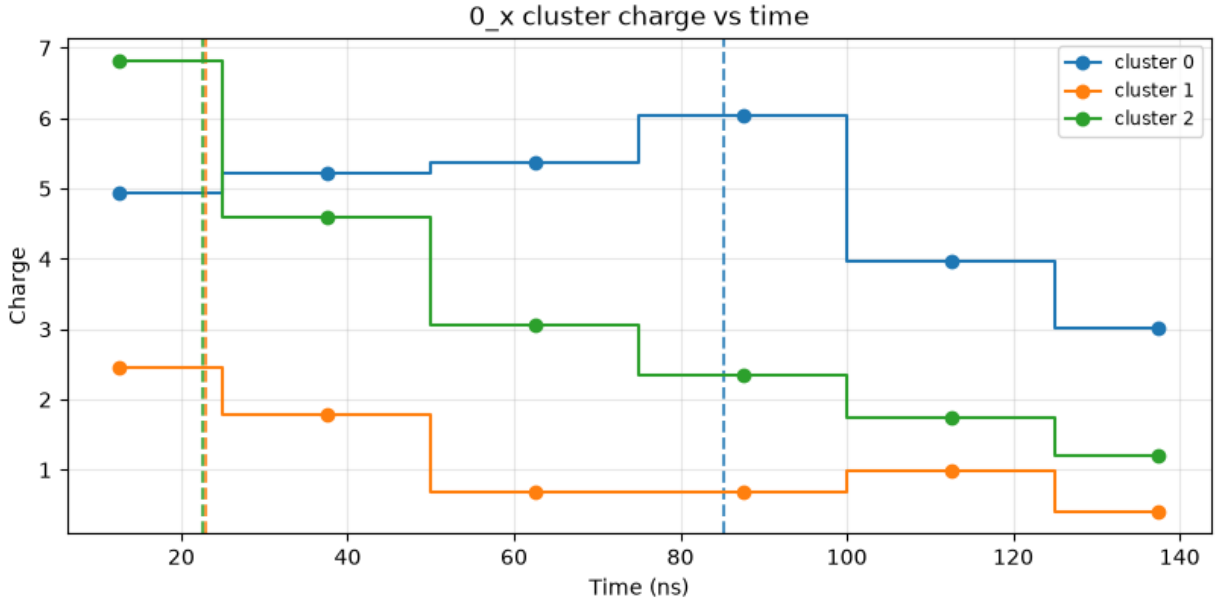
- Here, total charge is calculated using same method as before but for each time sample
  - Using same peak location and sigma determined earlier
- Dotted lines show weighted-average estimation of “cluster time”

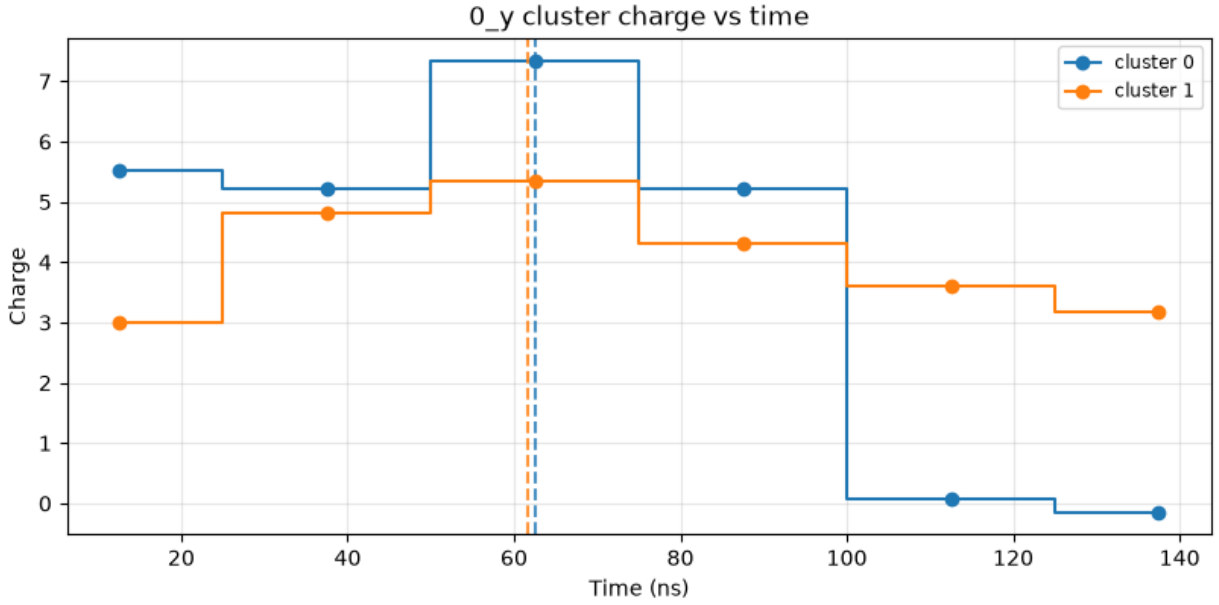
Possible ways to match X and Y clusters

- Maximize inner product or profiles
- Minimize mean squared error between profiles
- Find correlation between total or peak charge amplitude (based on how much charge is shared between X and Y strips)

Efficiency of matching X and Y clusters will be low because of multiple hit ambiguity

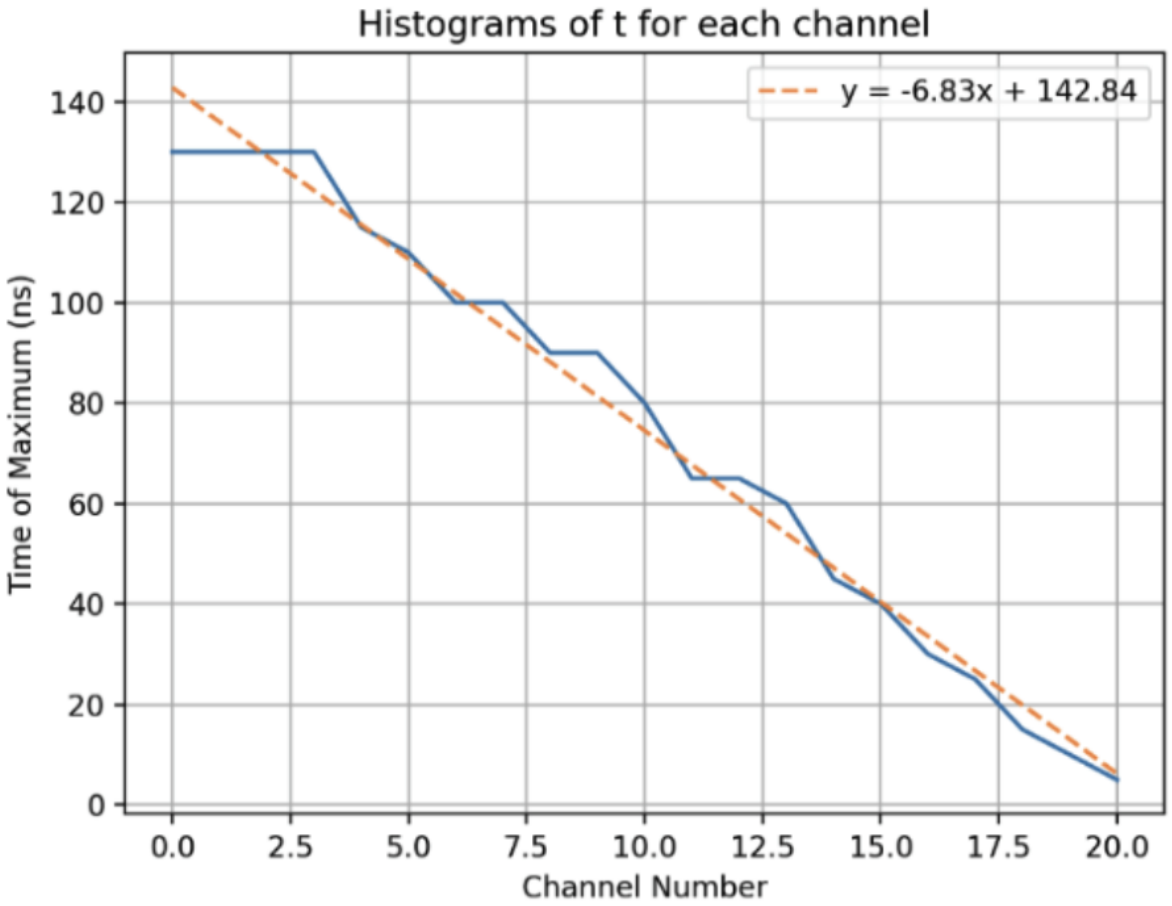
- Need to use information from other detectors





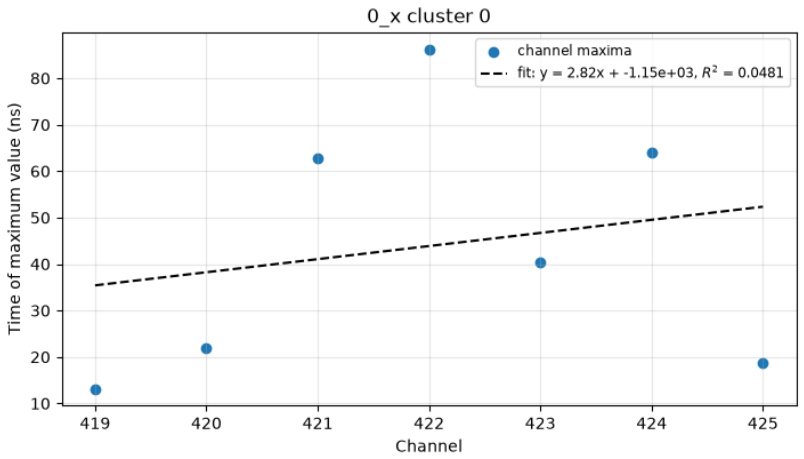
Time projection analysis

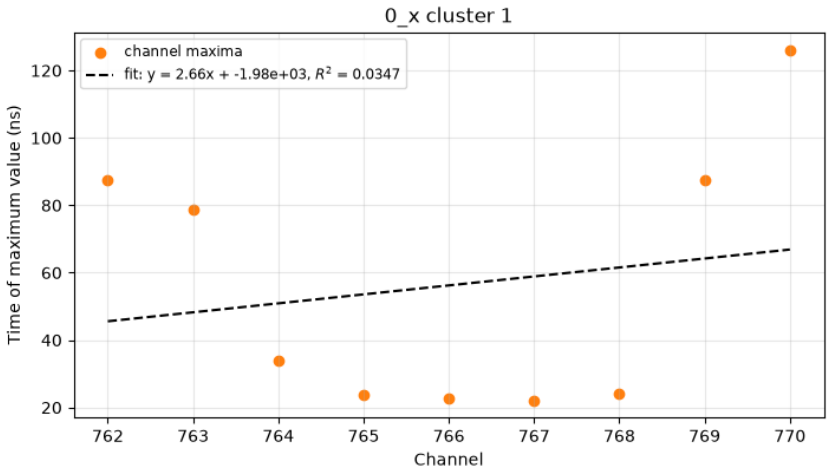
- Plot time of maximum charge on each channel in cluster
- Extract angle of incidence of particle track from slope
- We may not have enough time resolution
- More difficult in events with multiple tracks
  - Charge in each 1D cluster is projection of charge accumulated from potentially several 2D hits

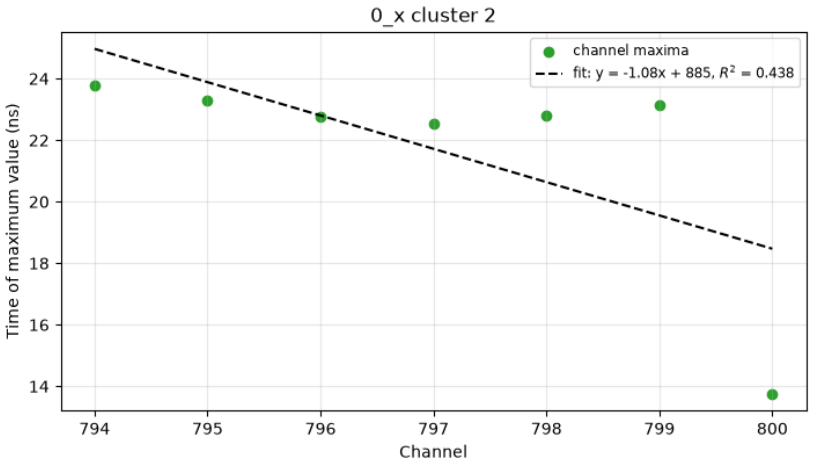


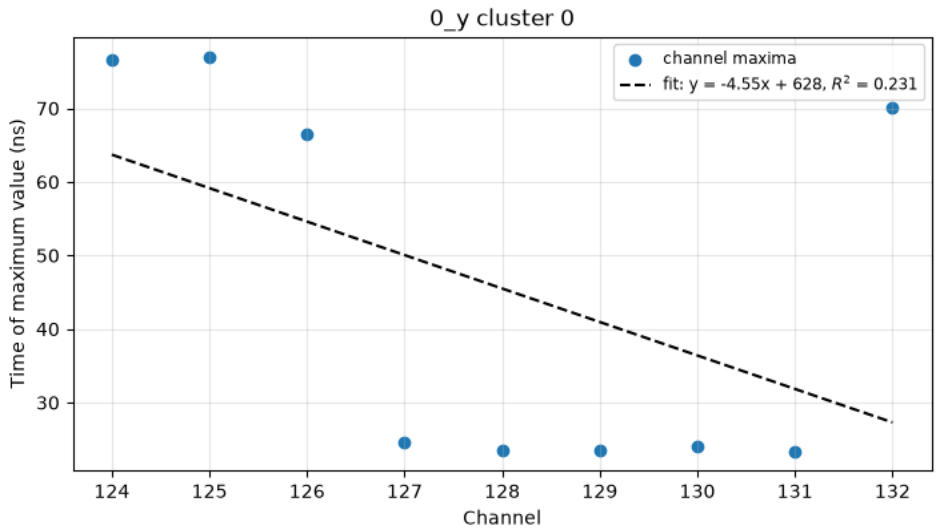
$$\text{Supposedly, } |\text{slope}| = (0.4\text{mm}) \cdot \frac{\cot\theta}{v_d} \sim 6.9$$

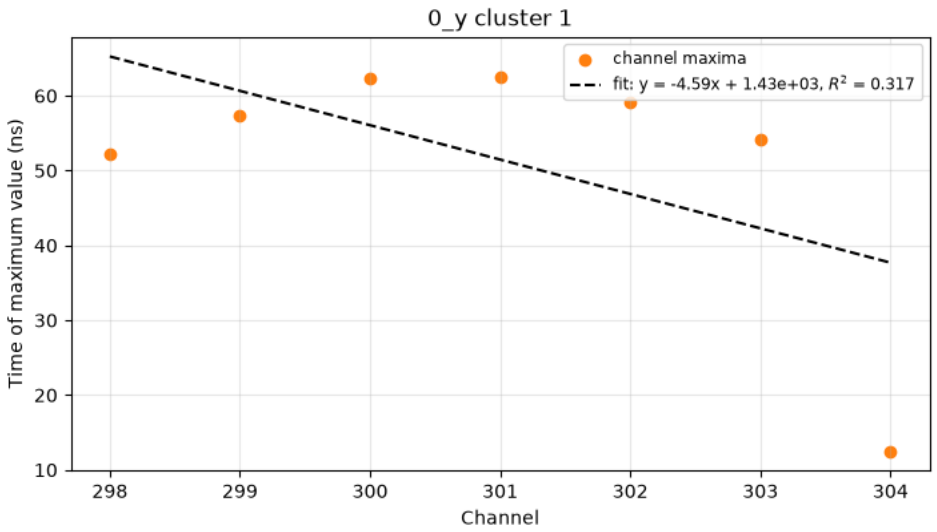
Time projection analysis











Cluster Analysis

Next Steps

- Improve estimation of width, charge, position, etc.
  - Find baseline and calculate height of signal above baseline
  - Calculate position with center-of-gravity method
- Test different ways of matching X and Y clusters
  - Error between charge versus time profiles
  - Charge/Peak amplitude – what fraction of charge is shared?
  - Similarity – inner product of charge profiles
- Look into extraction of angle information
  - See if we can calculate drift velocity

Notes

- Try this analysis on cleaner event with only one cluster per GEM
- Compare charge and timing with other GEM and with trigger hits